Project Initialization and Planning Phase

Date	26 June 2026
Team ID	-
Project Title	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution uses machine learning to predict credit card approval based on applicant details. It automates the evaluation process, improves prediction accuracy, reduces manual effort, and provides faster decision-making for both applicants and financial institutions.

Project Overview	
Objective	Develop a machine learning system that predicts whether a credit card application should be approved using applicant financial and personal information.
Scope	The system analyzes applicant data, predicts approval eligibility, and displays results through an easy-to-use web interface.
Problem Statement	
Description	Manual credit card approval is time-consuming, inconsistent, and prone to human error, resulting in delays and inaccurate decisions.
Impact	Slow approvals reduce customer satisfaction and increase operational workload for financial institutions

Proposed Solution	
Approach	Build a machine learning model that analyzes applicant details and predicts credit card approval eligibility automatically.

Key Features	• Automated approval prediction • Fast decision-making • Accurate ML-based analysis • Secure applicant data handling • Simple web interface

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU for model training and prediction	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for project files, datasets, and models	512 GB SSD
Software		
Frameworks	Backend framework	Flask
Libraries	Machine learning & data processing	Scikit-learn, Pandas, NumPy, Joblib
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Source, size, format	Application Record.csv Credit Record.csv Merged dataset (~36,000 applicants)